

CS 170 Java Tools—Summer '10

This summer in CS 170 we'll be using the **DrJava** Integrated Development Environment. You're free to use other tools but this is what you'll use on your programming exams, so you'll need to be familiar with it. The version of DrJava we'll be using has been *customized* for CS 170 to include built-in style checking and providing feedback on your programming assignments. This short document will show you how to download and configure it on your own personal computer.

Step 1: Got Java?

Before you can run DrJava, you first need to have **Java** on your computer. How you do that depends on what kind of computer you have. Here are the instructions.

- If you have a Mac (and you run Software Update regularly), *you're finished*. Java should be installed and correctly configured.
- If you are using Ubuntu Linux you should be able to install and configure Java using the shell command **sudo apt-get install sun-java6-jdk**
- If you are using Windows or another version of Linux, download the JDK (not the JRE) for your particular platform at: <http://java.sun.com/javase/downloads/index.jsp>



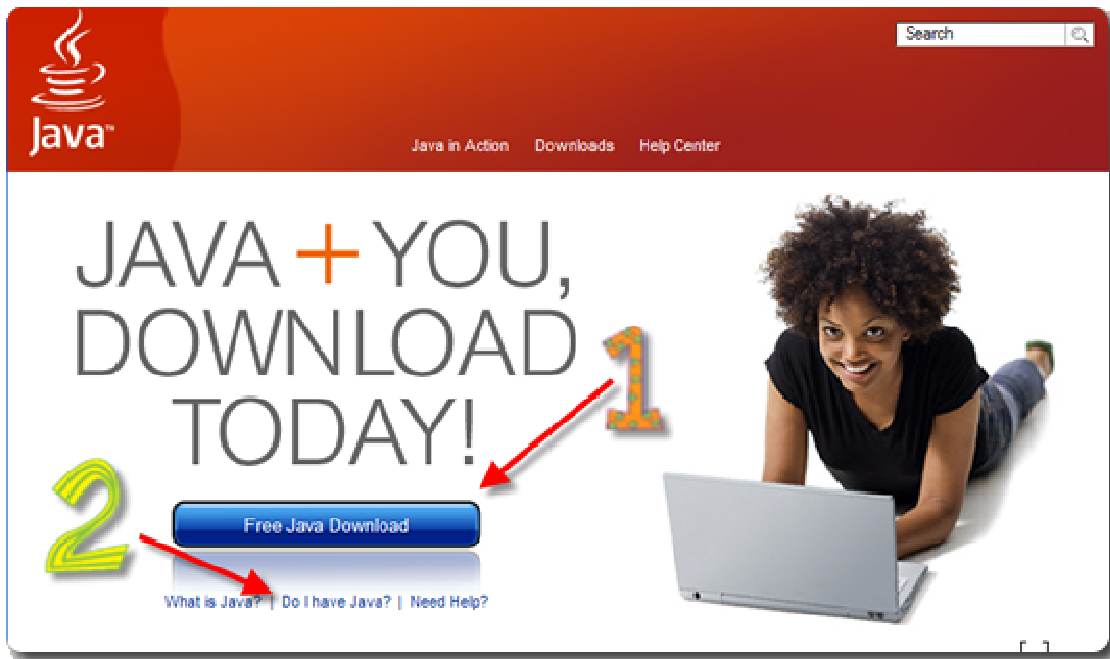
- Then, follow the installation instructions at:
<http://java.sun.com/javase/6/webnotes/install/index.html>

If you don't want to install the JDK (Windows and Linux only), you can install the JRE (Java Runtime Environment) instead, and use the Eclipse compiler which is built into DrJava. The next section of this document tells you how to do that. With the JRE only, you won't be able to generate Javadoc or use the Java debugger; neither of those things are really essential for this course. (You will learn how to use the DrJava debugger during one lesson in Week 5, but it isn't essential for any other part of the class.)

Note: if you are using Linux, you *can't* use the GNU version of Java, **gcj**. One way to check is to type **java -version** in a shell window. Even though **gcj** doesn't work, if you have **OpenJDK** already installed, that may work.

JRE Alternative

With Windows (and maybe Linux), you may get by just by installing a JRE (or Java Runtime Environment). You *actually* may already have one installed. You can try this first, if you like, and then install the JDK if this doesn't work for you. To install the JRE, just visit <http://java.com>

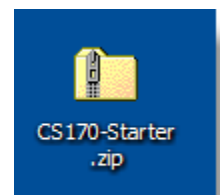


Click the link that says **Free Java Download** (1) and follow the instructions to install the program. If you want to see if you already have Java installed before you start, just click the hyperlink that says "Do I have Java installed" (2).

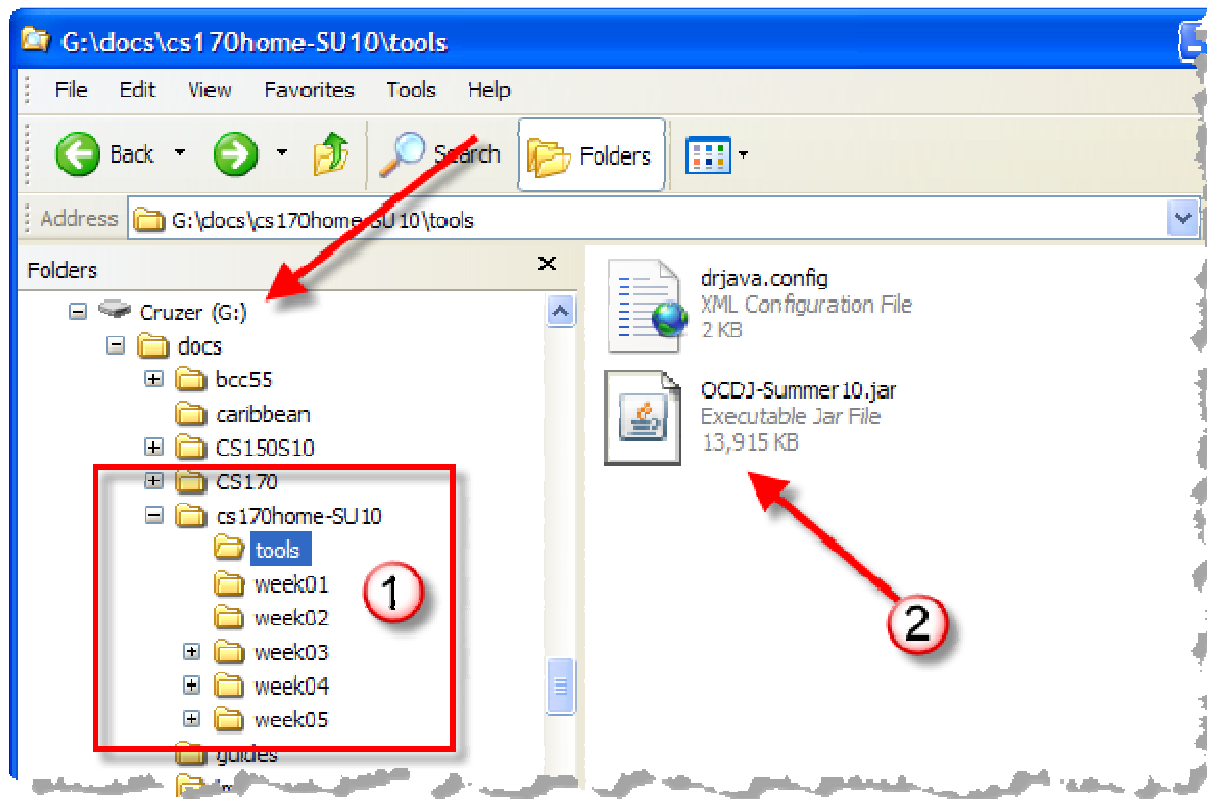
This works because the latest version of DrJava contains a built-in compiler (from the Eclipse IDE) that is activated if a JDK can't be found on your machine.

Step 2: Call the Doctor

Start by downloading the Zip file **cs170home-SU10.zip** from the syllabus link or from the class home page. You can put it on your desktop or in your documents folder. Once you've downloaded it, unzip it on your local machine. (Make sure that you actually *unzip* the file. Don't use the Windows feature that lets you treat the file as a compressed folder. If the zipped file looks like the picture here—with a zipper up the middle—then right-click it and choose Extract All from the context menu.)



Once you've unzipped **cs170home-SU10.zip** you find a folder named **cs170home-SU10**. This is the folder that contains DrJava and the folder you'll use for all your assignments this semester. Go ahead and drag it wherever you want to keep it. In the picture shown below, you can see that I've dragged the **cs170home-SU10** folder to a USB thumb drive named **Cruzer**, and placed it in my **docs** folder:



As you can see, the **cs170home-SU10** folder has a subfolder named **tools**, as well as folders for each week of the Summer coursework. Open the **tools** folder and double-click our customized version of DrJava, **OCDJ-Summer10.jar**

If all is well, you'll see the DrJava "splash screen" appear as DrJava loads for the first time. I've tried this out on Windows (through version 7) as well as on several different Macs.



Students have also used this on Linux, as long as you don't have the CompWiz window manager, which isn't compatible with DrJava.

Note: you will probably see an "update DrJava" dialog sometime during the semester. **Do NOT update** the customized version of DrJava, since it erases the features you'll want to use.

Troubleshooting Your Installation

DrJava doesn't start. If you have an "unarchiver" program like WinZip, it's possible for that program to "grab" the JAR file extension, so that when you double-click the file it gets unzipped instead. (This may happen in different version of Linux as well.) If that occurs, you need to open a

command shell window and use the command **java -jar OCDJ-Summer10.jar**. Of course, before you do this, make sure the you are "inside" the **tools** folder by typing **dir** (or **ls** on Linux), and making sure that **OCDJ-Summer10.jar** is actually resident in your local directory. On Linux you may need to make the jar file executable as well.

If you cannot get any of these to work, ask on the discussion board and get the rest of the class to give you a hand. Please do this early and not at the end of the week.

No compiler available. If you've tried the JRE only installation, when you start up DrJava on a machine without a JDK, and you see the message shown here when your program starts up, make sure that you are version 6 of the JRE. This message occurs with earlier versions of the JRE that cannot use the built-in copy of the Eclipse compiler. Note also that this only applies to Windows. This should not happen on a Mac.

